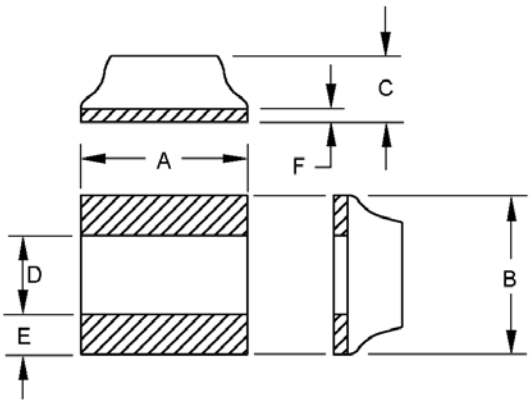
PERFORMANCE SPECIFICATION SHEET

COIL, RADIO FREQUENCY, CHIP, FIXED, CONFORMAL COATED, ESTABLISHED RELIABILITY, SURFACE MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-PRF-32560.



Dim.	Inches	Millimeters
A	.100 ± .010	2.54 ± .25
B	.100 ± .010	2.54 ± .25
C	.085 max.	2.16 max.
D	.030 min.	.762 min.
E	.015 min. 2 PL	.38 min.
F	.020 max. 2 PL	.51 max.

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Shape of curved portions optional provided dimensions A, B, and C are met.

FIGURE 1. Dimensions and Configuration.

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REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Material: Phenolic core (Inductance values 0.010 uH through 0.100 uH, inclusive); Ferrite core (0.12 uH through 27.0 uH, inclusive).

Weight: 0.5 gram maximum.

Operating temperature range: -55ºC to +125ºC.

Temperature rise (at 90ºC): 35ºC maximum. Test performed with coil mounted on SMD test substrate.

Maximum operating temperature: 125ºC.

Altitude: 70,000 feet maximum.

Dielectric Withstanding Voltage: In accordance with MIL-PRF-32560.

Barometric Pressure: In accordance with MIL-PRF-32560.

Insulation Resistance: In accordance with MIL-PRF-32560.

Points of test voltage application for Dielectric Withstanding Voltage, Barometric Pressure, and Insulation Resistance are between the terminals of the coil shorted together and the contact arm or assembly. The contact arm is made of conductive, moisture-resistant, resilient material and it shall cover the entire surface opposite the terminals, and shall be held firmly in place against this surface during testing.

Electrical characteristics (initial): See Table I.
Inductance and tolerance: See Table I.
Quality Factor (Q): See Table I.
Self-resonant frequency: See Table I. DC resistance: See Table I.

Terminal (Bond) Strength: 0.5 pound, when tested in accordance with MIL-STD-883, Method 2011, Test condition F. Test performed with coil mounted on SMD test substrate.

Solderability: In accordance with MIL-PRF-32560 with the following exceptions:

a. MIL-STD-202-208, test method B, or J-STD-002, Test Method S (except solder preforms may be used)

Resistance to solvents: Test is not applicable.

Overload: Test performed with coil mounted on SMD test substrate.

Low temperature storage: Test performed with coil mounted on SMD test substrate.

Vibration: In accordance with MIL-PRF-32560. Test performed with coil mounted on SMD test substrate.

Shock (Specified Pulse): In accordance with MIL-PRF-32560. Test performed with coil mounted on SMD test substrate.

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Life: Test performed with coil mounted on SMD test substrate.

Moisture Resistance: In accordance with MIL-PRF-32560.

SMD test substrate (where applicable): Material shall be made of a minimum of 96 percent alumina, or equivalent. The test substrate shall be prepared with metallized surface land areas.

Coil Mounting (where applicable): Test coils are to be soldered to the SMD test substrate's metallized areas only. Solder used shall be Sn63Pb37 solder or equivalent. The test substrate shall then be placed in or on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc.) with the temperature maintained at 260ºC +/- 5ºC, until the solder melts and reflows forming a homogenous solder connection.

Electrical characteristics (final): See Table II. For any Subgroup test requiring coils to be mounted to an SMD test substrate, the electrical characteristics (final) measurements are to be referenced to the electrical characteristics (initial) measurements determined after the test coil is mounted to the test substrate.

Part Marking: Coil marking is not applicable due to body size constraints.

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Table I Electrical characteristics (initial) and dash numbers.

Dash Number 1/	Inductance 4/ µH	Inductance Tolerances (%)	Q 4/ (min)	Test Frequency (MHz)	Self resonant frequency (MHz) (min)	DC resistance (ohms) (max)	2/ 3/ Current (mA) (max)
BR010***	0.010	±5,±10	60	50	2700	0.060	1270
BR012***	0.012	±5,±10	57	50	2450	0.069	1190
BR015***	0.015	±5,±10	55	50	2200	0.078	1110
BR018***	0.018	±5,±10	52	50	2000	0.093	1030
BR022***	0.022	±5,±10	50	50	1800	0.108	950
BR027***	0.027	±5,±10	49	50	1625	0.114	925
BR033***	0.033	±5,±10	48	50	1450	0.120	900
BR039***	0.039	±5,±10	45	50	1335	0.133	860
BR047***	0.047	±5,±10	42	50	1220	0.145	820
BR056***	0.056	±5,±10	39	50	1110	0.170	760
BR068***	0.068	±5,±10	36	50	1000	0.195	705
BR082***	0.082	±5,±10	34	50	915	0.212	675
BR10***	0.10	±5,±10	32	50	830	0.230	650
BR12***	0.12	±5,±10	30	25	750	0.125	880
BR15***	0.15	±5,±10	25	25	650	0.175	745
BR18***	0.18	±5,±10	25	25	550	0.200	695
BR22***	0.22	±5,±10	25	25	450	0.220	665
BR27***	0.27	±5,±10	25	25	375	0.230	650
BR33***	0.33	±5,±10	25	25	300	0.235	645
BR39***	0.39	±5,±10	22	25	235	0.240	635
BR47***	0.47	±5,±10	22	25	215	0.260	610
BR56***	0.56	±5,±10	22	25	195	0.278	590
BR68***	0.68	±5,±10	22	25	175	0.520	435
BR82***	0.82	±5,±10	22	25	160	0.530	430
B1R0***	1.0	±5,±10	22	25	145	0.540	425
B1R2***	1.2	±5,±10	22	7.9	130	0.740	360
B1R5***	1.5	±5,±10	22	7.9	115	0.840	340
B1R8***	1.8	±5,±10	22	7.9	105	0.920	325
B2R2***	2.2	±5,±10	22	7.9	85	1.00	310
B2R7***	2.7	±5,±10	24	7.9	77	1.15	290
B3R3***	3.3	±5,±10	24	7.9	70	1.40	260
B3R9***	3.9	±5,±10	24	7.9	68	1.55	250
B4R7***	4.7	±5,±10	24	7.9	60	1.80	230
B5R6***	5.6	±5,±10	22	7.9	55	2.00	220
B6R8***	6.8	±5,±10	22	7.9	50	2.20	210
B8R2***	8.2	±5,±10	22	7.9	48	2.50	195
B100***	10.0	±5,±10	24	7.9	40	3.45	165
B120***	12.0	±5,±10	25	2.5	35	3.80	160
B150***	15.0	±5,±10	25	2.5	30	5.60	135
B180***	18.0	±5,±10	25	2.5	28	5.80	130
B220***	22.0	±5,±10	25	2.5	25	6.40	125
B270***	27.0	±5,±10	25	2.5	22	6.90	120

1/ The complete dash number will include three (3) additional letters (indicated by ***). The first additional letter will indicate the inductance tolerance (e.g. J = ±5%, K = ±10%), the second additional letter will indicate the termination finish (e.g. A = Gold, B = Tin Lead, C=Palladium Silver), and the third additional letter will indicate the product level (e.g. M, P, R, S) and will be added to the end of the dash number.

2/ Maximum current allowed not to exceed the specified maximum temperature rise.

3/ These values are also the maximum incremental current applicable to ferrite cores only.

4/ Inductance and Q to be measured using HP4291A; HP 4194A or equivalent using TF16092; TF16034 or equivalent.

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Table II Electrical characteristics (final).

Inspection group	Allowable variation from initial measurements			
	Inductance (Percent)	DC resistance	Self-resonant frequency (Percent)	Q (Percent)
Qualification inspection				
Group II	± 5	±(3% +.001 ohm)	-10	-10
Group IIIA	± 5	----	---	-10
Group IV	± 5	±(3% +.001 ohm)	-8	-10
Group V	± 5	±(3% +.001 ohm)	-10	-10
Group B inspection				
Subgroup 1	± 5	±(3% +.001 ohm)	-8	-10
Subgroup 2	± 5	±(3% +.001 ohm)	-10	-10
Subgroup 3a	± 5	----	----	-10
Subgroup 4	± 5	±(3% +.001 ohm)	-8	-10

Part or Identifying Number (PIN): The part number shall be in the following form.

M32560/01-******* (dash number per Table I)

Referenced documents: In addition to MIL-PRF-32560, this document references:

MIL-STD-202-208, MIL-STD-883, J-STD-002

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

Review Activities:
Army – AR, CR4, MI
Navy – AS, CG, MC, OS
Air Force – 19, 99
NASA – NA
Other - MDA

(Project 5950-2017-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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